

Explainable Bangla Document Classification Using Attention-Based Deep Learning Models

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1 Literature Review

Rana et al. (2024) looked at how some models worked. These models were BanglaBERT, mBERT and XLM-RoBERTa. They used something called the Unified Bangla Multi-class Emotion Corpus to test these models. Hossain and his team were able to get good results with these models. They were able to understand different types of emotions. This study also created a standard to measure how well other models work on Bangla text. The study showed that the transformer architectures are really good at understanding the context of the text. However the texts they used for the study were mostly, from media. This means the results may not be as good if they used news articles. [1].

Hossain et al.(2023) looked at some models like BanglaBERT, mBERT and XLM-RoBERTa in 2023. They used these models with the Unified Bangla Multi-class Emotion Corpus. The study by Hossain et al did well with many types of emotions and created a standard dataset for classifying text in Bangla. The research by Hossain et al shows that transformer architectures are very good at understanding the context of the text. However the data they used is mostly from media so it may not work as well with formal news articles, like the ones you find in newspapers.[2].

Ahmad et al. (2022) explored several machine learning and deep learning approaches for Bangla news classification using the Potrika dataset. Their experiments showed that a FastText-based GRU model achieved the highest accuracy of 91.83%, outperforming traditional classifiers such as SVM and Logistic Regression. The study demonstrates the effectiveness of deep learning for Bangla news categorization. Nevertheless, automated topic labeling through LDA resulted in significantly lower classification performance. [3].

Roy et al. (2024) made a system to find news in Bangla. This system uses something called Bidirectional GRU. They used a lot of information around 50,000 pieces of data to test this system. The system was very good at finding news it was right about 99.16 percent of the time. The way this system is built helps it understand the context of what's being said from both sides of the conversation. Even though the system is very good it is so good that it makes us wonder if it is too perfect. If it will work in real life with real people and real news. The Bangla fake news detection system is what we are talking about here. This is the problem we are trying to solve with the Bidirectional GRU-based framework, for Bangla fake news detection. [4].

Rifa et al. (2024) introduced CineXDrama, a dual-stage transformer pipeline for relevance detection and sentiment analysis in Bangla entertainment reviews. The framework achieved 83.99% accuracy for relevance detection and 93.30% for sentiment classification. In addition, LIME was used to improve model interpretability and explainability. However, the relevance filtering stage may allow noisy data to propagate into the sentiment analysis phase. [5].

Mahmud et al. (2024) proposed a hybrid sentiment analysis framework that combines Bangla

Sentiment Polarity Score (BSPS) with BanglaBERT. The study demonstrated improved sentiment classification performance by effectively handling negation and intensity-related linguistic features. The integration of rule-based and transformer-based methods improved classification reliability. However, the lexicon component requires significant adaptation when applied to new domains. [6].

Islam et al. (2025) introduced BanglaSentNet in the year 2025. This is a kind of hybrid deep learning framework. It is used for -aspect sentiment analysis. The BanglaSentNet model uses a combination of LSTM, BiLSTM, GRU and BanglaBERT. It uses a strategy to combine these. The model got a good F1-score of 0.88. The team also used SHAP and attention visualization techniques. These techniques help make the model more transparent.. Even with these good things the BanglaSentNet model still has some problems. It is sensitive, to text that mixes Bangla and English which is called Bangla- code-mixed text [7].

Huang et al. (2022) proposed a Hierarchical Hybrid Neural Network with Multi-Head Attention (HHNN-MHA) for document classification. The model captures information at both word and sentence levels through a hierarchical structure and achieved strong performance on benchmark datasets. The attention mechanism improved feature extraction from long documents. However, the architecture requires considerable computational resources for training and deployment. [8].

Hussen et al. (2026) investigated explainable AI techniques for Bangla news classification. Their framework combined machine learning algorithms with LIME-based explanations to improve transparency and trustworthiness. Experimental results showed that explainable models can maintain competitive classification performance while providing useful insights into model decisions. However, the additional explanation stage increases inference time. [9].

Hassan et al. (2025) used something called XLM-RoBERTa to look at what people think about crime news in Bangladesh. The XLM-RoBERTa model was very good, at this. Got it right 97 percent of the time. It also used techniques to make it easier to understand how it worked. This study showed that a certain kind of computer model called transformer-based architectures can work well even when there is not a lot of information.. When people used slang or mixed languages the model did not do as well with XLM-RoBERTa. The XLM-RoBERTa model had trouble with this kind of language.[10].

Hasan et al. (2025) came up with a way to classify news titles in Bangla. This method is called BanglaBERT. It is semi-supervised. They used a technique where they gave some labels to the data and also tried to explain how the artificial intelligence works. This way they did not need to label all the data. They still got good results. When they tested this method they found that it was very accurate. It got 95.65 percent of the news titles right. The F1-score was 0.9544.. We do not know how well BanglaBERT works with the kind of language people use on social media, which is often informal. BanglaBERT is still not tested much on this type of content.[11].

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